

Deep Learning for Solving Dynamic Models

Session 4: Overlapping generations with DEQNs

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Based on Azinovic, Gaegauf & Scheidegger (2022, *IER* 63(4), 1471–1525) and Krueger & Kübler (2004, *JEDC*).

Roadmap

Part I: What is an OLG model?

Why overlapping generations; the cohort structure; the DEQN recipe in one line.

Part I½: The simplest OLG, two periods (Diamond, 1965)

One decision, one Euler equation, one closed form. Notebook 04_00.

Part II: The analytic six-agent model (Krueger & Kübler, 2004)

Euler equation and closed form derived; recursive equilibrium; cost function; network; validation. Notebook 04_01.

Part III: The 56-cohort benchmark of Azinovic, Gaegauf & Scheidegger (2022)

CRRA, hump-shaped labor, two assets, two constraints, persistent shocks. Notebook 04_02.

Learning objectives

By the end of this lecture you should be able to:

- ▶ Derive the Euler equation and the closed-form savings rate of a log-utility OLG economy, and say precisely which two assumptions make the closed form possible
- ▶ Construct a finite-dimensional overlapping-generations equilibrium with cohort-specific savings and lifecycle heterogeneity
- ▶ Derive equilibrium conditions and build the mean-squared Euler-residual loss for a multi-cohort OLG model
- ▶ Implement the DEQN recipe: model specification, equilibrium definition, cost function, network training, and validation
- ▶ Validate a learned solution against the closed-form equilibrium in the analytic Krueger–Kuebler model
- ▶ Scale the method from six cohorts to research benchmarks with capital adjustment costs, multiple assets, and binding constraints

Why Overlapping Generations?

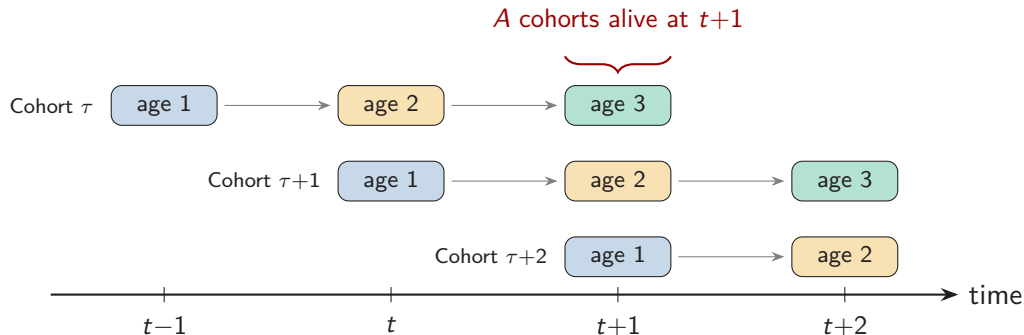
The economic content

- ▶ **Lifecycle:** agents are born, work, save, retire, die. A representative infinitely-lived household cannot speak to pensions, retirement, inheritance, or cohort-specific shocks.
- ▶ **Demographics:** at any t , many cohorts coexist, each with its own asset holdings and labor income.
- ▶ **Finite asset distribution:** with A cohorts, the cross-sectional wealth distribution is an A -vector, not an infinite-dimensional object. **Perfect for DEQNs:** state stays finite-dimensional with realistic heterogeneity.

Two canonical versions we solve today

1. $A = 6$ analytic model (Krueger & Kübler, 2004): closed form $\Rightarrow$ ground-truth validation.
2. $A = 56$ leading-example benchmark (Azinovic, Gaegauf & Scheidegger, 2022, §3): research-scale, no closed form.

OLG Structure: Overlapping Cohorts



At each period t we have A coexisting cohorts. A newborn enters (age 1); the oldest dies (age A).

Key observation: the number of types is finite (A), so the cross-sectional state is finite-dimensional.

The DEQN Recipe in One Line

From model to neural network (to be applied twice in today's lecture)

1. Write down the **economic model**: preferences, technology, constraints, shocks.
2. Define the **equilibrium**: choices + prices that satisfy optimization + market clearing.
3. Write the **equilibrium conditions** as a system $G(x, f(x)) = 0$.
4. Replace f by a **neural network** $\mathcal{N}_\theta$ and form the **cost function**
$$\mathcal{L}(\theta) = \frac{1}{|D|} \sum_j \|G(x_j, \mathcal{N}_\theta(x_j))\|^2.$$
5. Sample training states D , minimize $\mathcal{L}(\theta)$ by SGD.

The only thing that depends on the model

The **equilibrium conditions** G . Everything else, the network, the SGD loop, the sampling method, is generic. **So to solve a new model, write down its G .**

Today's deliverable: derive G explicitly for both the analytic and the benchmark OLG models.

Diamond (1965): Two Periods, One Decision

Households. Live two periods.

- ▶ **Young:** work (one unit), earn w_t ; consume c_t^y , save s_t in capital: $c_t^y + s_t = w_t$.
- ▶ **Old:** no labor; sell the capital, consume the proceeds: $c_{t+1}^o = r_{t+1} s_t$.
- ▶ Preferences $\ln c_t^y + \beta \mathbb{E}_t \ln c_{t+1}^o$, $\beta = 0.7$.

Firm. Same technology as later:

$\eta(z)K^\alpha L^{1-\alpha} + (1 - \delta(z))K$, $L = 1$. The old hold all the capital: $K_t = s_{t-1}$.

$$r_t = \alpha \eta(z_t) K_t^{\alpha-1} + 1 - \delta(z_t), \quad w_t = (1 - \alpha) \eta(z_t) K_t^\alpha.$$

State

(z_t, k_t) with $k_t = K_t$ the capital the old carry in. Prices are functions of it.

Shocks

The same four i.i.d. states (η, δ) as in Part II, probability $\frac{1}{4}$ each.

Two cohorts alive at once is the smallest overlapping structure there is.

The Euler Equation, Derived and Solved

Substitute both budget constraints: the young choose s to maximise

$$\ln(w_t - s) + \beta \mathbb{E}_t \ln(r_{t+1} s).$$

Step 1. $\ln(r_{t+1}s) = \ln r_{t+1} + \ln s$: the return and the choice **separate**.

Step 2. First-order condition:

$$\frac{1}{w_t - s} = \beta \mathbb{E}_t \left[\frac{r_{t+1}}{r_{t+1} s} \right] = \frac{\beta}{s} \quad \Rightarrow \quad s_t = \frac{\beta}{1 + \beta} w_t = 0.4118 w_t$$

Two observations

1. The return cancels. Under log utility the savings *rate* does not depend on the distribution of r_{t+1} , whether it is risky or safe, persistent or i.i.d.
2. The old have no wage, so today's wage is the household's entire lifetime resource. There is nothing to borrow against.

The DEQN for It

Unknown: one function $s = f(z, k)$.

Network: $\mathcal{N}_\theta(z, k) \rightarrow \hat{\sigma} \in (0, 1)$, a savings fraction of the wage, $\hat{s} = \hat{\sigma} w(z, k)$. So $0 \leq \hat{s} \leq w$ and $c^y = w - \hat{s} > 0$ by construction.

Residual (relative Euler error): tomorrow's capital is $K' = \hat{s}$, tomorrow's return $r'(z', K')$ follows for each of the four shocks, the old consume $c^{o'} = r' \hat{s}$, and

$$e(z, k) = \frac{(\beta \mathbb{E}[r'/c^{o'}])^{-1}}{w - \hat{s}} - 1.$$

Loss: mean of e^2 over a cloud of states that is re-simulated under the current policy after every gradient step.

What is missing so far

Tomorrow's policy does not enter: the old consume what they hold. With three or more cohorts the residual has to evaluate $\mathcal{N}_\theta$ at the next state, and that recursion through the network's own future behaviour is what a Deep Equilibrium Net handles.

The notebook computes r' and $c^{o'}$ explicitly although they cancel, so that the code carries over to the six-cohort case unchanged.

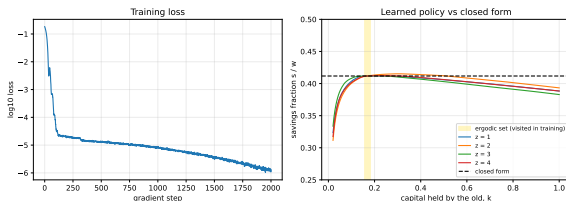
Finger Exercise: Two Periods

1. Part II will state a general formula for the savings rate of a cohort with $A - h$ periods still ahead: $\beta_h = \beta (1 - \beta^{A-h}) / (1 - \beta^{A-h+1})$. Evaluate it at $A = 2$, $h = 1$.
2. Suppose tomorrow's return r_{t+1} becomes *riskier* (same mean, more spread). What happens to s_t ? Why?
3. Now suppose the old *also* received a wage w_{t+1} . Would the young still save a fixed fraction of w_t ? Would they ever want to borrow?

3 minutes. Then open 04_00_OLG_Diamond_Warmup.ipynb: about one minute on a laptop. The network should report $\hat{\sigma} = 0.4118$ at every state the economy visits, and drift away from it elsewhere, which is why training uses a simulated cloud.

Solution: Two Periods

1. $\beta_1 = \beta(1 - \beta)/(1 - \beta^2) = \beta/(1 + \beta)$: the two-period result is the $A = 2$ case of the general formula.
2. Nothing changes. $\mathbb{E} \ln(r's) = \mathbb{E} \ln r' + \ln s$, so the spread of r' only shifts a constant that the first-order condition does not see. (With CRRA, $\gamma > 1$, the young would save more, a precautionary motive, and the closed form is lost.)
3. No, and yes. A future wage is a second lifetime resource; the young would want to borrow against it when w_t is low, and saving would depend on (w_t, w_{t+1}, r_{t+1}) jointly. This is the 56-cohort benchmark of Part III in miniature.



II.1 Setup: the Analytic Model

- ▶ Time discrete: $t = 0, 1, 2, \dots, \infty$.
- ▶ Agents live for $A = 6$ periods.
- ▶ **One representative household per cohort**; a new one is born every period.
- ▶ **No uncertainty about lifetime**: all agents live exactly A periods.
- ▶ Exogenous aggregate shock $z_t \in \{1, 2, 3, 4\}$ drawn **i.i.d.** with transition $\pi_{ss'} = 0.25$.
- ▶ **Log utility** $u(c) = \ln c$, chosen because it gives a **closed-form** equilibrium (Krueger & Kübler, 2004).
- ▶ **Labor**: only the youngest cohort works: $\ell^1 = 1$, $\ell^h = 0$ for $h \geq 2$.

Shocks: $z_t = (\eta_t, \delta_t)$

z	$\eta(z)$	$\delta(z)$	π
1	0.95	0.5	0.25
2	1.05	0.5	0.25
3	0.95	0.9	0.25
4	1.05	0.9	0.25

Calibration

$\alpha = 0.3$, $\beta = 0.7$, $\gamma = 1$ (log); $\underline{a} = 0$.

II.2 Households: Preferences and the Savings Technology

Agents supply their labor **inelastically** for a market wage w_t .

An agent **currently of age h** alive at t maximises remaining time-separable discounted expected lifetime utility ($\beta < 1$):

$$\mathbb{E}_t \left[\sum_{i=0}^{A-s} \beta^i \ln(c_{t+i}^{h+i}) \right] \quad (1)$$

Households can save a unit of consumption to obtain a unit of capital good next period, denoted a_t^h .

The savings become **capital in the next period**:

$$a_t^h = k_{t+1}^{h+1}, \quad \forall t, \forall h \in \{1, \dots, A-1\}. \quad (2)$$

II.3 Households: Constraints, Budget, Boundary

Households **cannot die with debt**. Borrowing is bounded:

$$a_t^h \geq \underline{a}. \quad (3)$$

In the analytic model we set $\underline{a} = 0$: **no borrowing** for any cohort.

At time t , households sell their capital to the firm at market price $r_t > 0$. The per-period **budget constraint** of cohort h reads

$$c_t^h + a_t^h = r_t k_t^h + \ell^h w_t. \quad (4)$$

Boundary conditions. Newborns enter with no inheritance, and the oldest cohort dies without saving anything:

$$k_t^1 = 0, \quad a_t^A = 0.$$

II.4 Firms and Markets

There is a single representative firm with **Cobb–Douglas** production.

TFP η and depreciation δ depend on the exogenous shock z alone (see II.1 table).

Each period, **after the shock has realized**, the firm buys capital and hires labor to maximize profits, taking prices as given.

The stochastic production function is

$$f(K, L, z) = \eta(z) K^\alpha L^{1-\alpha} + K(1 - \delta(z)).$$

There are **competitive spot markets** for the consumption good, capital, and labor.

In the analytic model $L_t = \sum_{h=1}^A \ell^h = 1$ (only age-1 works), so the firm problem reduces to $\max_{K \geq 0} \eta(z) K^\alpha - (r_t - (1 - \delta(z)))K$.

II.5 Competitive Equilibrium: Definition

Definition 1 (competitive equilibrium)

Given initial conditions z_0 , $\{k_0^h\}_{h=1}^{A-1}$, a competitive equilibrium is a collection of choices for households $\{(c_t^h, a_t^h)_{h=1}^A\}_{t=0}^\infty$, for the representative firm $(K_t, L_t)_{t=0}^\infty$, and prices $(r_t, w_t)_{t=0}^\infty$, such that:

1. Given $(r_t, w_t)_{t \geq 0}$, household choices maximize the lifetime utility (1) subject to the savings-to-capital identity (2), the borrowing limit (3), and the per-period budget (4):

$$a_t^h = k_{t+1}^h; \quad a_t^h \geq \underline{a}; \quad c_t^h + a_t^h = r_t k_t^h + \ell^h w_t.$$

2. Given (r_t, w_t) , the firm maximises profits:

$$(K_t, L_t) \in \arg \max_{K, L \geq 0} f(K, L, z_t) - r_t K - w_t L.$$

3. **All markets clear:** for all t , $L_t = \sum_{h=1}^A \ell^h$, $K_t = \sum_{h=1}^A k_t^h$.

II.6 Equilibrium Conditions

Firm FOCs (closed form from Cobb–Douglas):

$$w_t = (1 - \alpha) \eta(z_t) K_t^\alpha L_t^{-\alpha}, \quad r_t = \alpha \eta(z_t) K_t^{\alpha-1} L_t^{1-\alpha} + (1 - \delta(z_t)).$$

Household optimality for any cohort of age $h \in \{1, \dots, A - 1\}$ (FOC in c_t^h gives $\nu_t^h = u'(c_t^h)$; FOC in a_t^h gives the standard **interior** Euler equation):

$$\underbrace{u'(c_t^h)}_{= 1/c_t^h \text{ since } u=\ln} = \beta \mathbb{E}[u'(c_{t+1}^{h+1}) r_{t+1}]$$

Terminal cohort. Age A : consume everything, save nothing.

Why no KKT multiplier? No cohort has future labor income (only the newborn works, once), so there is nothing to borrow against and $a_t^h \geq 0$ never binds: $\lambda_t^h \equiv 0$. Part III re-introduces multipliers, since a hump-shaped wage profile gives the young a reason to borrow.

II.6a Deriving the Euler Equation

Cohort h at date t chooses $\{c, a\}$ for its remaining $A - h + 1$ periods; ν is attached to the budget constraint (4), $\lambda \geq 0$ to the borrowing limit:

$$\mathcal{L} = \mathbb{E}_t \sum_{i=0}^{A-h} \beta^i \left[\ln c_{t+i}^{h+i} + \nu_{t+i}^{h+i} (r_{t+i} k_{t+i}^{h+i} + \ell^{h+i} w_{t+i} - c_{t+i}^{h+i} - a_{t+i}^{h+i}) + \lambda_{t+i}^{h+i} a_{t+i}^{h+i} \right], \quad k_{t+i+1}^{h+i+1} = a_{t+i}^{h+i}.$$

- FOC in c_t^h : $\nu_t^h = 1/c_t^h$, the marginal utility of cash.
- FOC in a_t^h : $-\nu_t^h + \lambda_t^h + \beta \mathbb{E}_t [\nu_{t+1}^{h+1} r_{t+1}] = 0$, since a_t^h becomes k_{t+1}^{h+1} and earns r_{t+1} .
- Complementary slackness: $\lambda_t^h \geq 0$, $a_t^h \geq 0$, $\lambda_t^h a_t^h = 0$.

$$\boxed{\frac{1}{c_t^h} = \beta \mathbb{E}_t \left[\frac{r_{t+1}}{c_{t+1}^{h+1}} \right] + \lambda_t^h} \quad a_t^A = 0, \quad c_t^A = r_t k_t^A.$$

With $\lambda_t^h = 0$ this is the interior Euler equation of II.6. Part III keeps λ and adds a second multiplier.

II.6b Why This Model Has a Closed Form, and Why the Benchmark Does Not

Two assumptions

1. **Log utility.** $\mathbb{E} \ln(r' a) = \mathbb{E} \ln r' + \ln a$, so the savings rate does not depend on the return distribution: income and substitution effects of a return shock cancel. Persistent shocks would change nothing here; i.i.d. shocks only keep the aggregate dynamics simple.
2. **No future labor income.** Only the newborn works, once. Cash-on-hand $W_t^h = r_t k_t^h + \ell^h w_t$ is the household's entire state, there is nothing to borrow against, and $\lambda \equiv 0$.

What the 56-cohort benchmark drops

- ▶ CRRA with $\gamma = 2$: the return no longer separates, and a riskier r' raises saving.
- ▶ Hump-shaped ℓ^h : the young have wages ahead of them and want to borrow, so the constraint binds, and so does a collateral constraint on bonds.

No closed form; multipliers in the network; a numerical solver is needed.

II.6c Backward Induction to β_h

Write W for cash-on-hand and $V_h(W)$ for the value of a cohort of age h .

- **Age A :** no tomorrow. $c = W$, $V_A(W) = \ln W$.
- **Age $A - 1$:** $\max_a \ln(W - a) + \beta \mathbb{E} \ln(r' a)$. FOC $\frac{1}{W-a} = \frac{\beta}{a} \Rightarrow a = \frac{\beta}{1+\beta} W$,
 $V_{A-1}(W) = (1 + \beta) \ln W + \text{const.}$
- **Age $A - 2$:** $\max_a \ln(W - a) + \beta(1 + \beta) \mathbb{E} \ln(r' a) \Rightarrow a = \frac{\beta + \beta^2}{1 + \beta + \beta^2} W$,
 $V_{A-2}(W) = (1 + \beta + \beta^2) \ln W + \text{const.}$
- **Age h , by induction:**

$$\beta_h = \frac{a}{W} = \frac{\beta + \beta^2 + \dots + \beta^{A-h}}{1 + \beta + \dots + \beta^{A-h}} = \beta \frac{1 - \beta^{A-h}}{1 - \beta^{A-h+1}}, \quad h = 1, \dots, A - 1.$$

The constant absorbs $\mathbb{E} \ln r'$ at every step, which is why the return distribution never reaches the decision. W is the inc_t^h of the budget constraint (4), so the sigmoid savings-fraction head of II.9 has the same shape as the closed form.

II.7 The DEQN Cost Function

Explicit cost function

$$\mathcal{L}_{D_{\text{train}}}(\theta) := \frac{1}{|D_{\text{train}}|} \frac{1}{A-1} \sum_{x_j \in D_{\text{train}}} \sum_{i=1}^{A-1} (e_{\text{REE}}^i(x_j))^2$$

Relative Euler-equation residual (cohort i):

$$e_{\text{REE}}^i(x_j) := \frac{\left(u'\right)^{-1}\left(\beta \mathbb{E}\left[r\left(\hat{x}_{j,+}\right) u'\left(\hat{c}^{i+1}\left(\hat{x}_{j,+}\right)\right)\right]\right)}{\hat{c}^i\left(x_j\right)} - 1.$$

Next state and expectation. $\hat{x}_{j,+} = (z_+, 0, \hat{a}^{[1:A-1]}(x_j))$, the newborn entering with $k^1 = 0$; $\mathbb{E}$ is the exact four-term sum over z_+ .

No market-clearing residual. Next period's capital is defined as $K_{t+1} = \sum_h \hat{a}^h(x_j)$, so the capital market clears by construction. Part III needs one for bonds.

II.8a From a Sequence to a Function

Definition 1 describes an equilibrium as sequences $\{c_t^h, a_t^h, r_t, w_t\}_{t \geq 0}$. A neural network represents a function, not a sequence. The link is a sufficient state.

$(z_t, \mathbf{k}_t)$ is all a household needs to know

- ▶ Today's prices: $K_t = \sum_h k_t^h$ and z_t pin down (r_t, w_t) , hence every cohort's cash-on-hand.
- ▶ Tomorrow's prices: if behaviour is the same in the same state, $(z_{t+1}, \mathbf{k}_{t+1})$ depends only on $(z_t, \mathbf{k}_t)$, the shock, and that behaviour.

We therefore look for a **recursive** equilibrium: time-invariant policies $a^h(z, \mathbf{k})$ with the law of motion $\mathbf{k}' = (0, a^1(z, \mathbf{k}), \dots, a^{A-1}(z, \mathbf{k}))$. Existence and stationarity are assumed; in Part II the closed form is such a function, so the assumption is verified.

The DEQN literature calls this object a functional rational expectations equilibrium: one function whose equilibrium residuals vanish at every state. The next slide writes its signature.

II.8 Functional Rational Expectations Equilibrium (FREE)

FREE: one function that solves the equilibrium conditions at every state

Signature: $f : \{1, 2, 3, 4\} \times \mathbb{R}^A \longrightarrow \mathbb{R}^{A-1}$, input = state $(z_t, \mathbf{k}_t)$, output = savings policies.

Output decomposition ($A - 1$ scalars):

$$f\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \end{bmatrix}\right) = \underbrace{[a_t^1, a_t^2, \dots, a_t^{A-1}]^\top}_{\text{capital-investment policies}}.$$

Such that, for all cohorts $h = 1, \dots, A - 1$, the (interior) Euler equation holds:

$$\beta \mathbb{E} \left[\frac{r_{t+1} u'(c_{t+1}^{h+1})}{u'(c_t^h)} \right] - 1 = 0 \quad (\text{Euler equation}).$$

Stack these $A - 1$ conditions as $G([z_t, \mathbf{k}_t], f([z_t, \mathbf{k}_t])) = \mathbf{0}$.

II.9 Deep Equilibrium Net: Architecture (Analytic)

$$\mathcal{N}_\theta : \{1, 2, 3, 4\} \times \mathbb{R}^A \rightarrow \mathbb{R}^{A-1}$$

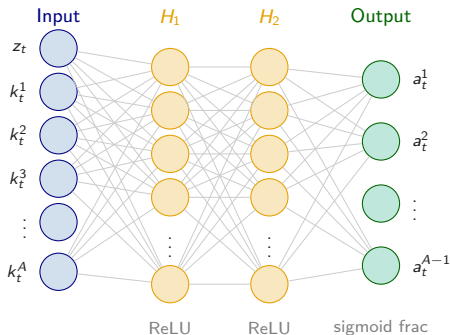
$$\mathcal{N}_\theta\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \end{bmatrix}\right) = \begin{bmatrix} a_t^1 \\ a_t^2 \\ \vdots \\ a_t^{A-1} \end{bmatrix}$$

such that

$$G\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \end{bmatrix}, \mathcal{N}_\theta\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \end{bmatrix}\right)\right) \approx \mathbf{0}.$$

Sigmoid savings-fraction output $\Rightarrow$

$0 \leq a_t^h \leq \text{inc}_t^h$ **by construction**, so $a_t^h \geq 0$
(borrowing constraint) and $c_t^h \geq 0$ both hold
at every iteration without an explicit multiplier.



2 hidden layers $\times$ (100, 50) units; single forward pass.

Companion Notebook Hand-over

Open notebook `04_01_OLG_Analytic_DEQN_persistent.ipynb` (the persistent-simulation primary; the exogenous-sampling companion is `04_04_OLG_Analytic_DEQN_exogenous.ipynb`; if you skipped it, `04_00` is the two-period warm-up with the same structure in sixty lines). It implements the model, equilibrium conditions, loss, FREE map, architecture, training, and validation covered in Part II.

Cell-to-slide map

Deck slide	Notebook section
II.1–II.6 Model and equilibrium	§1–3 parameters, state, network/policy
II.7–II.9 Loss, FREE, architecture	§4 <code>compute_cost()</code> + training-data construction
II.10–II.13 Training/validation	§5–8 mini-batches, diagnostics, policy stability, closed-form checks

II.10 Sampling and the Training Loop

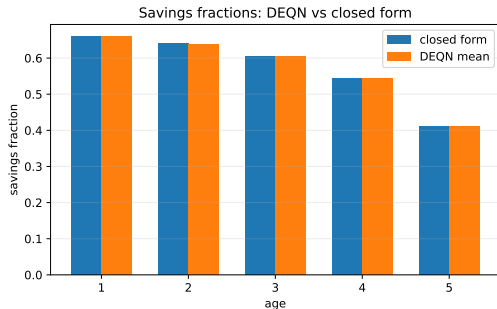
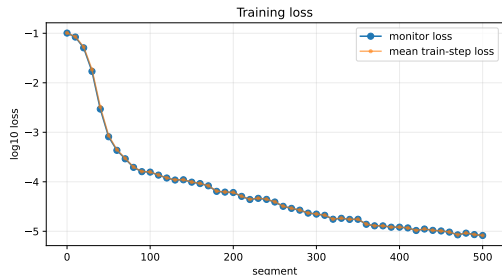
Cloud method (Azinovic, Gaegauf & Scheidegger, 2022)

1. **Initialisation.** Sample a cloud of initial states $D_0 = \{x_j^0\}_{j=1}^N$; initialise θ^0 .
2. **Episode k .**
 - (a) **Simulate.** Draw fresh shocks $\{z_{+,j}\}$ from π ; roll the cloud forward one period using the **current** policy $\mathcal{N}_{\theta^k}$: $x_j^{k+1} = (z_{+,j}, 0, \mathcal{N}_{\theta^k}^{[1:A-1]}(x_j^k))$.
 - (b) **Mini-batch SGD.** Take steps Adam steps on $\mathcal{L}_{D_k}(\theta)$ using mini-batches from D_k .
3. **Stop** when $\log_{10} \mathcal{L}$ falls below the target.

Definitions. **Episode** = one simulate-then-train pass. **Epoch** = one pass through the full cloud.

Sampling from the ergodic set of the current policy makes the cost function focus on states the economy actually visits.

II.11 Training Results: Analytic Model (teaching-mode run)



Reading the plots

Left: monitor loss and mean train-step loss per segment, log₁₀ scale; the cloud is re-simulated each segment.

Right: mean savings fraction by cohort next to the closed form β_h . From 04_01, RUN_MODE = "teaching", about five minutes on a laptop.

Finger Exercise: Closed-Form Savings Rates

Setup. With $u = \ln$, cohort h 's Euler equation reads $1/c_t^h = \beta \mathbb{E}_t[r_{t+1}/c_{t+1}^{h+1}]$ and the budget constraint is $c_t^h + a_t^h = \text{inc}_t^h$ where $\text{inc}_t^h = r_t k_t^h + \ell^h w_t$.

Claim (closed-form log-utility OLG savings rate; the validation target used by Krueger & Kübler, 2004). With log utility and 100% consumption at age A , the savings rate $\beta_h := a_t^h / \text{inc}_t^h$ of cohort h depends only on age:

$$\beta_h = \beta \frac{1 - \beta^{A-h}}{1 - \beta^{A-h+1}}, \quad h = 1, \dots, A-1.$$

Task (3 minutes). Compute $\beta_1, \beta_2, \dots, \beta_5$ with $\beta = 0.7$. Interpret:

- What happens as cohorts approach retirement?
- Why are younger cohorts saving more as a **rate**?

Solution: Closed-Form Savings Rates

$$\beta_h = 0.7 \cdot \frac{1 - 0.7^{6-h}}{1 - 0.7^{7-h}}$$

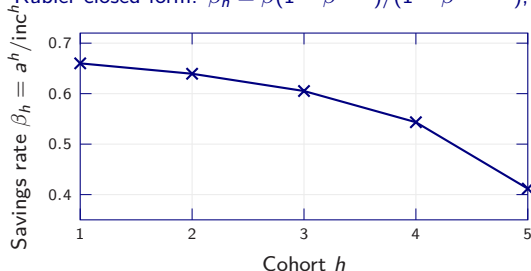
Cohort h	Periods remaining ($A - h$)	Savings rate β_h
1	5	$0.7 \cdot (1 - 0.7^5)/(1 - 0.7^6) \approx 0.660$
2	4	$0.7 \cdot (1 - 0.7^4)/(1 - 0.7^5) \approx 0.639$
3	3	$0.7 \cdot (1 - 0.7^3)/(1 - 0.7^4) \approx 0.605$
4	2	$0.7 \cdot (1 - 0.7^2)/(1 - 0.7^3) \approx 0.543$
5	1	$0.7 \cdot (1 - 0.7)/(1 - 0.7^2) \approx 0.412$
6	0	0 (consumes everything)

Interpretation

- ▶ Savings rate **monotonically declines** with age: as the horizon shrinks, saving less is optimal.
- ▶ Youngest cohort saves $\sim 66\%$ of its income, longest horizon to enjoy the returns.
- ▶ At A : terminal boundary $a^A = 0$, consume everything.

II.13 DEQN vs. Analytical Solution

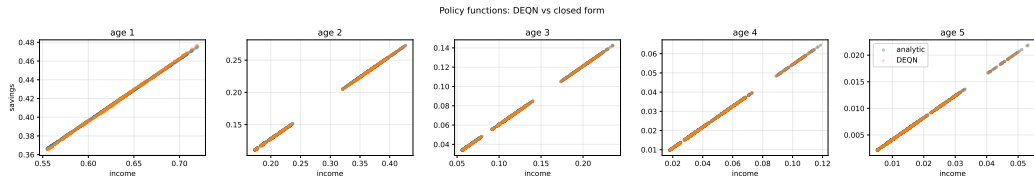
Krueger-Kübler closed form: $\beta_h = \beta(1 - \beta^{A-h})/(1 - \beta^{A-h+1})$, $\beta = 0.7$, $A = 6$



Validation strategy

The points are the exact closed form. The DEQN should reproduce them cohort by cohort; the next slide checks it state by state. Production target in AGS (2022): mean relative Euler error below 10^{-3} .

II.13' Policy Functions State by State



One panel per cohort: savings a^h against cash-on-hand inc^h on the simulated ergodic cloud. The closed form is a line through the origin with slope β_h ; the DEQN should lie on it at every income level.

The ergodic gross return is about 0.69, below one because $\delta \in \{0.5, 0.9\}$, and the young still save two thirds of their wage: capital is the only store of value in this economy.

II. Take-away and Hands-on

What we just did

1. OLG model with 6 cohorts, log utility, 4 i.i.d. shocks (II.1–II.4).
2. Equilibrium definition + conditions (II.5–II.6).
3. Conditions $\rightarrow$ **cost function** (II.7).
4. Equilibrium map $f \rightarrow$ neural network $\mathcal{N}_\theta$ (II.8–II.9).
5. Trained by sampling + SGD (II.10–II.11), **validated** against the closed form (II.13).

Hands-on

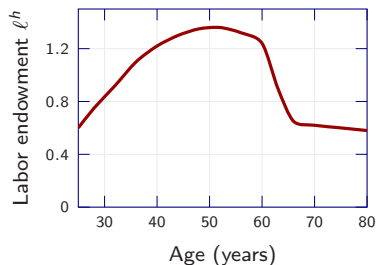
Open `04_01_OLG_Analytic_DEQN_persistent.ipynb`. Set `RUN_MODE = "teaching"` (~ 5 min) for the classroom run; "smoke" (the default) is a ~ 30 -s sanity check; "production" chases AGS Table 3 accuracy. Both notebooks expose a `SAMPLING_MODE = "simulation" / "exogenous"` switch (the persistent primary uses "simulation", the companion `04_04_OLG_Analytic_DEQN_exogenous.ipynb` uses "exogenous"); the model and loss are identical, only the training cloud differs.

III.1 Setup: the Leading Example of AGS (2022)

Same recipe as Part II. What changes:

- ▶ Agents live for $A = 56$ periods (ages 25–80); one period = one year.
- ▶ CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$ with $\gamma = 2$ (not log).
- ▶ **Hump-shaped labor endowment** ℓ^h (Kotlikoff-style; right panel).
- ▶ **Persistent** Markov shocks on (η, δ) (not i.i.d.).
- ▶ **Two assets**: capital k^h and bonds b^h .
- ▶ **Two constraints**: borrowing $k'^h \geq 0$ and collateral $k'^h + \kappa b'^h \geq 0$.
- ▶ Capital adjustment costs $\Psi^h = \frac{\zeta}{2}(k'^h - rk^h)^2$.

Stylised lifecycle labor (schematic)



Schematic shape only, rises through working years, drops sharply at retirement, low but nonzero after. See **Brumm, Kübler & Scheidegger (2017)** for the exact ℓ^h used in the notebook; AGS (2022) reuse this calibration as part of their benchmark.

III.2 Households: Preferences and Savings Technology

An agent of age h alive at t maximises

$$\mathbb{E}_t \left[\sum_{i=0}^{A-s} \beta^i u(c_{t+i}^{h+i}) \right], \quad u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \quad \gamma = 2. \quad (1')$$

Two saving instruments (instead of one):

- **Capital** k_{t+1}^{h+1} : yields gross return r_{t+1} next period; incurs an **adjustment cost**
 $\psi_t^h = \frac{\zeta}{2} (k_{t+1}^{h+1} - r_t k_t^h)^2$.
- **Bonds** b_{t+1}^{h+1} : traded at price p_t today, pay one unit of consumption next period. **Zero net supply**: $\sum_{h=1}^A b_t^h = 0$.

Savings become next-period asset holdings ($\forall h \in \{1, \dots, A-1\}$):

$$k_{t+1}^{h+1} = a_{t,k}^h, \quad b_{t+1}^{h+1} = a_{t,b}^h. \quad (2')$$

III.3 Households: Constraints, Budget, Boundary

Two constraints on the capital–bonds pair (k'^h, b'^h) :

$$k'^h \geq 0 \quad (\text{borrowing constraint}), \quad k'^h + \kappa b'^h \geq 0 \quad (\text{collateral constraint}). \quad (3')$$

Budget constraint of cohort h (bonds enter at price p_t on the left-hand side, pay off on the right):

$$c_t^h + k_{t+1}^{h+1} + p_t b_{t+1}^{h+1} + \psi_t^h = r_t k_t^h + b_t^h + \ell_t^h w_t. \quad (4')$$

Boundary conditions (born and die without assets):

$$k_t^1 = 0, \quad b_t^1 = 0; \quad k_{t+1}^{A+1} = 0, \quad b_{t+1}^{A+1} = 0.$$

Bond parameter $\kappa \geq 0$ governs how much bond-short position capital collateralises; $\kappa = 0$ is a **hard** short-sale constraint on bonds.

III.4 Firms and Markets: Adding the Bond Market

Representative firm. Same Cobb–Douglas as Part II:

$$f(K, L, z) = \eta(z) K^\alpha L^{1-\alpha} + K(1 - \delta(z)).$$

Labor now aggregates over the hump-shaped profile: $L_t = \sum_{h=1}^A \ell^h$ (a constant).

Markets cleared in equilibrium.

1. Labor: $L_t = \sum_{h=1}^A \ell^h$.
2. Capital: $K_t = \sum_{h=1}^A k_t^h$.
3. Bonds (new): $0 = \sum_{h=1}^A b_t^h$. (Zero net supply.)

Aggregate shocks. $z_t \in \{1, \dots, 4\}$ as before, but transition π^z is **persistent** (not i.i.d.): (η, δ) have their own Markov chains, tensor-producted.

III.5 Competitive Equilibrium: Definition

Definition 1' (competitive equilibrium, benchmark)

Given initial conditions $z_0, \{k_0^h, b_0^h\}_{h=1}^{A-1}$, a competitive equilibrium is a collection of choices $\{(c_t^h, k_{t+1}^h, b_{t+1}^h)_{h=1}^{A-1}\}_{t=0}^{\infty}$, firm demands $(K_t, L_t)_{t \geq 0}$, and prices $(r_t, w_t, p_t)_{t \geq 0}$, such that:

1. Households maximize (1') subject to (2')–(4').
2. Firm maximises profits, same closed-form FOCs as Part II.
3. All three markets clear every period: capital, labor, bonds.

Relative to Part II: one new price (p_t), one new market-clearing condition (bonds), one new asset class per cohort in the choice vector. Everything else is the same.

III.6 Equilibrium Conditions (Benchmark)

Firm FOCs. Same closed form as Part II.

Two Euler equations per cohort: for $h = 1, \dots, A - 1$ (with the marginal adjustment-cost wedge $\mathcal{A}_t^h := 1 + \zeta(k_t'^h - r_t k_t^h)$):

$$\text{(capital)} \quad u'(c_t^h) \mathcal{A}_t^h = \beta \mathbb{E}[u'(c_{t+1}^{h+1}) r_{t+1} \mathcal{A}_{t+1}^{h+1}] + \lambda_b^h + \mu^h,$$

$$\text{(bonds)} \quad u'(c_t^h) \cdot p_t = \beta \mathbb{E}[u'(c_{t+1}^{h+1})] + \kappa \mu^h.$$

Two KKT triples per cohort.

$$\lambda_b^h \geq 0, \quad k'^h \geq 0, \quad \lambda_b^h \cdot k'^h = 0; \quad \mu^h \geq 0, \quad k'^h + \kappa b'^h \geq 0, \quad \mu^h \cdot (k'^h + \kappa b'^h) = 0.$$

Terminal cohort. Age A : consume everything.

The wedge $\mathcal{A}'$ on the right: next period's adjustment cost $\frac{\zeta}{2}(k'' - r'k')^2$ depends on k' , with derivative $-r'\zeta(k'' - r'k')$, so the marginal return to k' is $r'\mathcal{A}'$. The bond equation with $\mu = 0$ is $p = \beta \mathbb{E}[u'(c')]/u'(c)$, the price of a riskless claim; a binding collateral constraint adds $\kappa\mu/u'(c)$. Per state: $4(A - 1)$ per-agent residuals + 1 bond-market residual = $4(A - 1) + 1$.

III.7 The DEQN Cost Function (Benchmark)

Five loss components, one cost function

$$\mathcal{L}_{D_{\text{train}}}(\theta) = \frac{1}{|D_{\text{train}}|} \frac{1}{4(A-1)+1} \sum_{x_j \in D_{\text{train}}} \left[\sum_{h=1}^{A-1} R^h(x_j)^2 + (e_{\text{MC},b}(x_j))^2 \right],$$

$$\text{where } R^h(x_j)^2 := (e_{\text{REE},k}^h)^2 + (e_{\text{REE},b}^h)^2 + (e_{\text{KKT},b}^h)^2 + (e_{\text{KKT},c}^h)^2.$$

Component map.

- ▶ $e_{\text{REE},k}^h, e_{\text{REE},b}^h$: relative residuals of the capital and bond Euler equations.
- ▶ $e_{\text{KKT},b}^h = \lambda_b^h \cdot k'^h$: borrowing KKT.
- ▶ $e_{\text{KKT},c}^h = \mu^h \cdot (k'^h + \kappa b'^h)$: collateral KKT.
- ▶ $e_{\text{MC},b} = \sum_{h=1}^{A-1} \hat{b}'^h$: bond market clearing (**chosen** next-period bonds, not the current state).

With $A = 56$ this is $4 \times 55 + 1 = 221$ loss terms per training state. The structure is the same as Part II: squared equilibrium residuals.

III.8 Functional Rational Expectations Equilibrium

FREE for the benchmark

$$f : \{1, \dots, 4\} \times \mathbb{R}^A \times \mathbb{R}^A \longrightarrow \mathbb{R}^{4(A-1)+1}$$

$$f\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \\ \mathbf{b}_t \end{bmatrix}\right) = \begin{bmatrix} \hat{k}_t'^1, \dots, \hat{k}_t'^{A-1} \\ \hat{\lambda}_{b,t}^1, \dots, \hat{\lambda}_{b,t}^{A-1} \\ \hat{b}_t'^1, \dots, \hat{b}_t'^{A-1} \\ \hat{\mu}_t^1, \dots, \hat{\mu}_t^{A-1} \\ \hat{p}_t \end{bmatrix}^\top$$

Output blocks: capital savings; borrowing multipliers; bond holdings $\hat{b}^h$ (transformed to satisfy the collateral floor $\hat{k}^h + \kappa \hat{b}^h \geq 0$); collateral multipliers; equilibrium bond price (one scalar).

Such that $G([z_t, \mathbf{k}_t, \mathbf{b}_t], f(\cdot)) = \mathbf{0}$, where G stacks the $4(A-1) + 1$ equations of III.6.

With $A = 56$: minimal state dimension 113; the notebook input is 240 after engineered aggregates and transition probabilities; output dimension $4 \times 55 + 1 = 221$.

III.9 Architecture: Benchmark

$\mathcal{N}_\theta : \mathbb{R}^{240} \rightarrow \mathbb{R}^{221}$, two ReLU hidden layers of 128 (teaching) or 1000 (production) units, Adam.

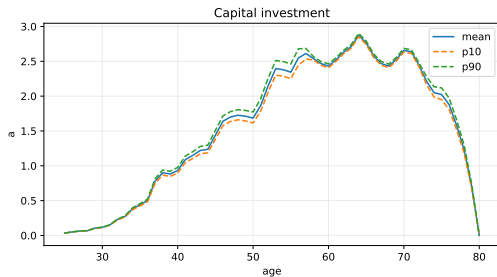
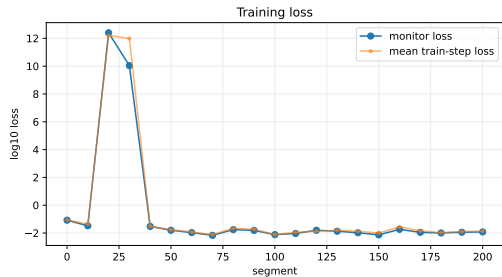
Output heads, from 04_02

- ▶ $\hat{k}'^h$ and $\hat{q}^h$: a bounded exponential times a state-dependent baseline, $a_{\text{base}} \exp(1.5 \tanh z)$, equal to the baseline at zero raw output.
- ▶ $\hat{\lambda}_b^h, \hat{\mu}^h$: softplus, so non-negative. $\hat{p}$: β times a bounded exponential.
- ▶ $\hat{b}'^h = (\hat{q}^h - \hat{k}'^h)/\kappa$.

What this head cannot do

With it $\hat{k}'^h \geq 0.0045$ and $\hat{q}^h > 0$ for every network output, so neither constraint can bind: the multipliers go to zero and III.12 reports binding fractions of zero. A head that reaches zero (CONSTRAINT_HEAD in 04_02) does not train at the teaching preset and is left as the extension.

III.10 Training: Loss and Lifecycle Capital (teaching-mode run)

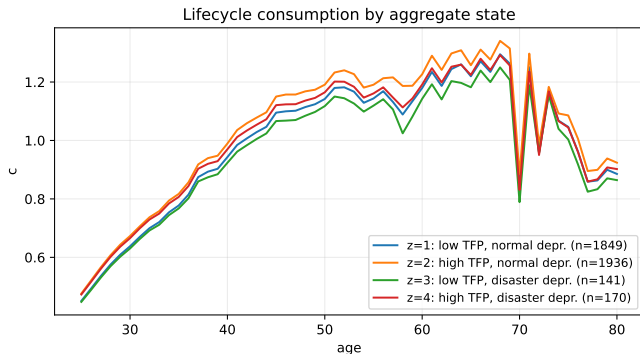


Reading the plots

Left: loss per segment; each segment re-simulates the cloud under the current policy.

Right: capital saving by age on the ergodic cloud, mean and 10/90 percentiles. Teaching preset, 128^2 units, 200 segments; the production run in AGS (2022) tightens the residuals but not the shape.

III.11 Consumption Profiles by Aggregate State

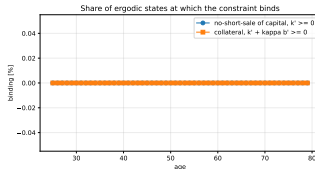


Mean lifecycle consumption on the ergodic cloud, one curve per aggregate state $z = (\eta, \delta)$, from 04_02 §8. Teaching run: mean consumption is highest in state 2 (high TFP, normal depreciation), then states 4, 1 and 3; the TFP gap outweighs the depreciation gap. The ordering is measured on the ergodic cloud, not conjectured.

III.12 Reading the Diagnostics

How do we know the network solved the equilibrium?

1. **Loss level and Euler errors by age:** mean $|e_{REE,k}^h|$, $|e_{REE,b}^h|$ around 10^{-3} in production.
2. **Bond market clearing:** $|e_{MC,b}|$ small on the ergodic path.
3. **Binding fractions by age** (figure): share of ergodic states at which $\hat{k}''^h$ is at zero with $\hat{\lambda}_b^h > 0$, and likewise for the collateral pair. With the default head both are zero by construction (III.9); the diagnostic is what shows it.
4. **Policy drift on a fixed anchor cloud:** the run is time-invariant once `policy_drift_rms` and `_max` are below tolerance.
5. **Lifecycle profiles** on a long simulated path (III.10, III.11).



Hands-on: 04_02_OLG_Benchmark_DEQN_persistent.ipynb; 04_05 is the exogenous-sampling ablation.

Teaching runtime about ten minutes on a laptop.

Summary

What we accomplished

1. OLG models: a finite-dimensional family of heterogeneous-agent models, suited to DEQNs.
2. Applied the DEQN recipe **three times**:
 - ▶ **Part I½**: Diamond's two-period model, one Euler equation, solved by hand and by the network.
 - ▶ **Part II**: the 6-agent Krueger–Kübler (2004) analytic model, validated against the closed form ($\lesssim 10^{-3}$ mean Euler error).
 - ▶ **Part III**: the 56-cohort benchmark of AGS (2022): CRRA, hump-shaped labor, persistent shocks, two assets, two constraints, adjustment costs.
3. **Same recipe** every time: model, equilibrium conditions, cost, recursive equilibrium, network, training, validation.

One line to remember

Writing down a DEQN = writing down the equilibrium conditions G . Everything else (network, SGD, sampling) is boilerplate.

References and Further Reading

- ▶ Azinovic, M., Gaegauf, L., & Scheidegger, S. (2022). *Deep equilibrium nets*. International Economic Review 63(4), 1471–1525.
[04_02_OLG_Benchmark_DEQN_persistent.ipynb implements §3 of this paper].
- ▶ Krueger, D., & Kübler, F. (2004). *Computing equilibrium in OLG models with stochastic production*. Journal of Economic Dynamics & Control 28(7), 1411–1436.
[04_01_OLG_Analytic_DEQN_persistent.ipynb implements their 6-agent analytic model].
- ▶ Diamond, P. A. (1965). *National debt in a neoclassical growth model*. American Economic Review 55(5), 1126–1150. [the two-period model of Part I½; 04_00_OLG_Diamond_Warmup.ipynb].
- ▶ Fischer, A. (1992). *A special Newton-type optimization method*. Optimization 24, 269–284. [FB-complementarity function, used in the **IRBC chapter (Ch. 3)**; mentioned in this lecture only as an alternative to the product-form KKT residuals adopted here.]
- ▶ Code: <https://github.com/sischei/DeepEquilibriumNets>.

Bridge: one more dimension to add

What we just did. Cohort structure changed the size of the residual system, one Euler equation per generation, derived once from the Lagrangian and stacked into one loss, but not the training principle. The two-period model gave the idea, the six-cohort model a closed form to validate against, and the 56-cohort benchmark added what breaks the closed form: CRRA, future wages, and the borrowing and collateral constraints the young actually hit, encoded as product-form KKT residuals.

Next (Session 5): the international real business cycle model. OLG stacked equations along the *age* dimension. IRBC stacks them along a *country* dimension instead: N Euler equations plus a world resource constraint on a $2N$ -dimensional state. Same loss, same training loop, and the point where a tensor-product grid is no longer an option.

Bring with you. The architecture search and loss balancing from Session 3: IRBC is the first model in this course whose residual components genuinely live on different scales.